# Ch. 15 Solutions

Principles of Microeconomics

August 3, 2026

## Quick Quiz

1. (c) – Because there are many sellers all of whom offer the same product, firms are price takers in a perfectly competitive market.

2. (d) – Marginal revenue is the price of the good. In a perfectly competitive industry, one firm increasing its quantity supplied doesn't change the price because all firms are price takers. Therefore, marginal revenue stays the same.

   Total revenue equals price times quantity, so it's proportional to the quantity sold. If quantity increases by 10%, total revenue increases by 10%.

3. (b) – If they chose a quantity for which $MC < MR$, they could increase their profit by producing one more unit because one more unit brings in more revenue than it costs them to produce. If they chose a quantity for which $MC > MR$, they could increase their profit by producing one fewer unit because their last unit brought in less revenue than it cost them to produce. Combined, that means they should produce the quantity at which $MC = MR$.

4. (d) – The profit-maximizing quantity for a firm is the quantity at which marginal cost equals marginal revenue. In a competitive market, marginal revenue equals price, so the profit-maximizing quantity is the quantity at which price equals marginal cost. That means the marginal cost curve plots the profit-maximizing quantity for a firm at any given price. Since firms are profit-maximizers, that means the marginal cost curve plots the quantity supplied by a firm at any given price.

   The caveat to that analysis is that a firm might be better off not producing at all. If a firm shuts down in the short run, it won't make any revenue, but it will save all of its variable costs. That means a firm is better off not producing in the short run if its revenue is less than its variable costs. With a little bit of math, we can show that that's the same thing as saying the firm is better off not producing in the short run if the market price is less than average variable cost:

   $$\begin{aligned} TR &< VC \\ \frac{TR}{Q} &< \frac{VC}{Q} \\ \frac{PQ}{Q} &< AVC \\ P &< AVC \end{aligned}$$

   Therefore, in the short run, if price is less than average variable cost, a firm will shut down, but if price is greater than average variable cost, a firm will produce the quantity given by the marginal cost curve. In other words, the firm's short-run supply curve is the marginal cost curve above its average variable cost curve.

5. (a) – Firms produce the quantity at which marginal cost equals price, so if marginal cost is between average variable cost and average total cost, that means price is between average variable cost and average total cost.

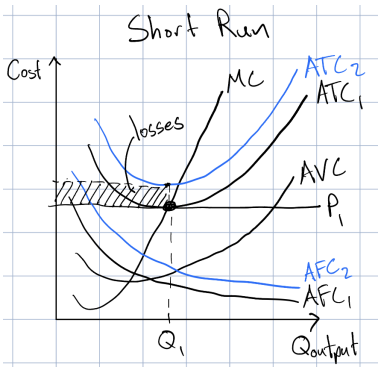
If a firm shuts down in the short run, it will lose all of its revenue, but it will save all of its variable costs (it's fixed costs are sunk, so those are irrecoverable in the short run). Therefore, the firm will keep producing if its revenue is greater than its variable costs. That's the same thing as saying the firm will keep producing in the short run if price is greater than average variable cost. Since that's the case here, the firm continues to produce in the short run.

If a firm shuts down in the long run, it will lose all of its revenue, but it will save all of its costs (variable and fixed because the fixed costs become recoverable in the long run). Therefore, the firm will exit if its revenue is less than its costs. That's the same thing as saying the firm will exit in the long run if price is less than average total cost. Since that's the case here, the firm exits in the long run.

6. (d) – Firms will produce the quantity at which $P = MC$ because they're profit maximizers. They'll enter or exit the market until profits are zero which happens when $P = ATC$.

7. (c) – Firms will produce the quantity at which $P = MC$ because they're profit maximizers. If new firms want to enter, profits must be positive which means $P > ATC$.

8. (c) – In the short run, the tax is an increase in fixed costs and total costs. Marginal cost and average variable cost don't change, so firms' quantity produced doesn't change in the short run. That means the number of pretzels consumed stays the same in the short run. In the long run, firms exit the market because ATC shifts up and profit becomes negative. When firms exit, the quantity produced decreases, so the number of pretzels consumed decreases.




## Questions for Review

1. The two main characteristics are:

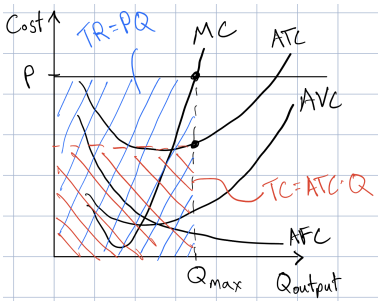
   - the market has many buyers and sellers, and
   - all of the sellers produce identical goods/services.

   The third characteristic that economists only sometimes assume is:

   - firms can freely enter and exit the market.

2. A firm's revenue is the total amount of money that it brings in. A firm's profit is the total amount of money that it brings in minus its costs of production. Firms try to maximize profit.

3. A firm chooses the quantity at which marginal cost equals marginal revenue. If the firm chose a quantity where marginal cost was less than marginal revenue, then producing one more unit would raise costs less than it would raise revenue, so producing one more unit would raise profit. As a result, the firm would want to produce more. If the firm chose a quantity where marginal cost was greater than marginal revenue, then producing one fewer unit would lower costs less than it would lower revenue, so producing one fewer unit would raise profit. As a result, they'd want to produce less. The point where they're producing the perfect profit-maximizing amount is right where marginal cost equals marginal revenue.




4. If a firm shuts down, it loses all of its revenue, but it saves all of its variable costs. Recall that shutting down is a short-run decision, and fixed costs are sunk in the short run, so the firm can't save its fixed costs by shutting down. Therefore, a firm will only shut down if its revenue is less than its variable costs. With a little bit of math, we can show that that's the same thing as saying a firm will only shut down if the market price is less than the firm's average variable costs:

$$\begin{aligned} TR &< VC \\ \frac{TR}{Q} &< \frac{VC}{Q} \\ \frac{PQ}{Q} &< AVC \\ P &< AVC \end{aligned}$$

5. If a firm exits, it loses all of its revenue, but it saves all of its costs. Recall that exiting the market is a long-run decision, and fixed costs become variable in the long run, so the firm can save all of its fixed costs as well as its variable costs. Therefore, a firm will exit if its revenue is less than its total cost. With a little bit of math, we can show that that's the same thing as saying a firm will exit if the market price is less than the firm's average total cost:

$$\begin{aligned} TR &< TC \\ \frac{TR}{Q} &< \frac{TC}{Q} \\ \frac{PQ}{Q} &< ATC \\ P &< ATC \end{aligned}$$

6. A competitive firm's price equals its marginal cost in both the short run and the long run. Firms are profit-maximizers, so if they choose to produce, they will produce the quantity at which marginal revenue equals marginal cost. In a competitive market, firms are price takers, so marginal revenue is equal to the market price. That means firms will always produce the quantity at which marginal cost equals market price. The only caveat to that statement is that sometimes firms will choose not to produce at all. In the short run, a firm will shut down if price is less than its average variable cost. In the long run, a firm will exit if price is less than average total cost.

7. A competitive firm's price equals the minimum of its average total cost in the long run. In the short run, it's difficult for firms to enter or exit the market, so profits could be positive or negative. In either case, price wouldn't equal average total cost. In the long run, firms can easily enter or exit, so profits tend towards zero. If profits are zero, that means price equals average total cost. We know that firms produce the quantity at which price equals marginal cost, so in the long run, price equals average total cost and marginal cost. The only place where average total cost and marginal cost are equal is at the minimum of average total cost. Therefore, in the long run, price equals the minimum of average total cost.

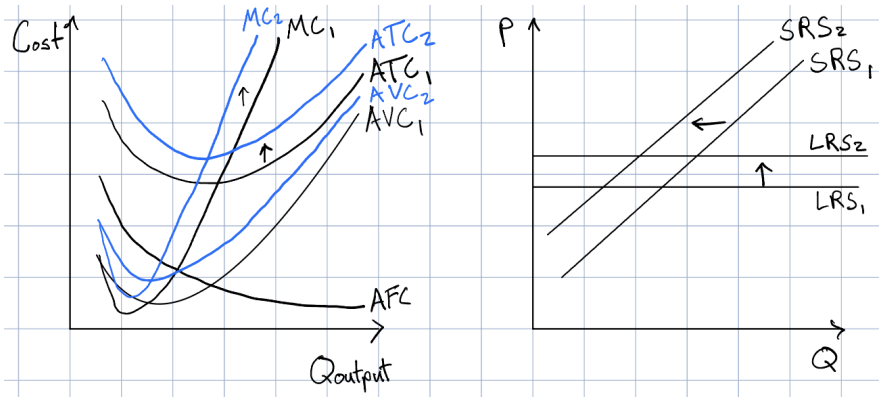
8. Market supply curves are typically more elastic in the long run. In the long run, firms can enter and exit the market. As a result, quantity supplied is more responsive to a change in price. Therefore, supply is more elastic.

## Problems and Applications

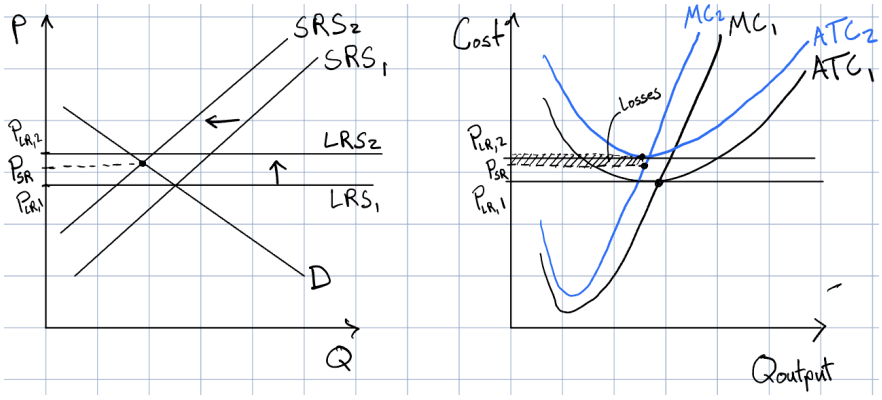
1. (a) An increase in the price of an input increases marginal cost, variable cost, and total cost. As a result, the marginal cost, average variable cost, and average total cost curves shift up.

   The short-run market supply curve is the sum of all the individual firms' supply curves. The individual firms' supply curves are their marginal cost curves. Since each firm's marginal cost curve shifts up (a.k.a. left), the short run market supply curve must shift left as well.

   The long run market supply curve is horizontal at the minimum average total cost of production. Since the average total cost curve shifts up, the minimum average total cost of production increases. As a result, the long run market supply curve shifts up.




   (b) In the short run, profits and the number of boat makers depend on the elasticity of demand. As I drew it below, demand is elastic enough that the new short run price, $P_{SR}$, is below the new minimum average total cost, $P_{LR,2}$. In that case, profits become negative, and boat makers exit the market. If demand were more inelastic, $P_{SR}$ could be above $P_{LR,2}$. In that case, profits become positive and boat makers will enter the market.




2. Leah's variable cost per day is

$$\$280 - \$30 = \$250.$$

   Therefore, Leah's average variable cost is

$$\$250/10 = \$25.$$

   That's less than her price of \$27, so Leah produces in the short run.

Leah's average total cost is

$$\$280/10 = \$28.$$

That's greater than her price of \$27, so Leah exits in the long run.

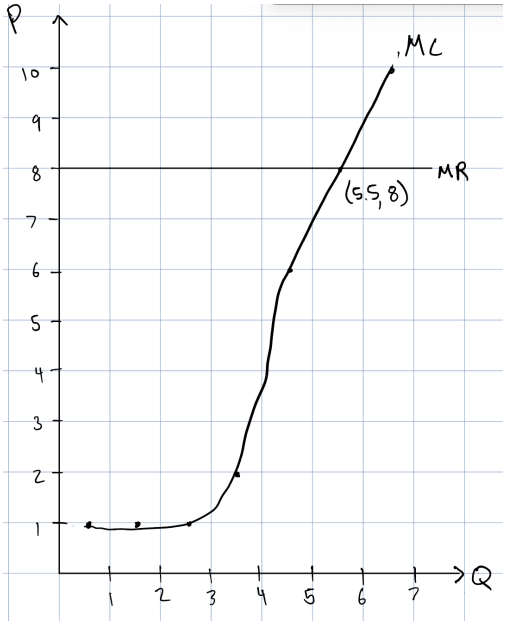
3. (a) Profit is total revenue minus total cost.

| **Quantity** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| **Profit** | -8 | -1 | 6 | 13 | 19 | 21 | 21 | 19 |

The firm should produce 5 or 6 units to maximize profit.

(b) Marginal revenue is the increase in revenue from increasing production by one unit. Marginal cost is the increase in cost from increasing production by one unit.

| **Quantity** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| **MR** | – | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| **MC** | – | 1 | 1 | 1 | 2 | 6 | 8 | 10 |




They intersect at a quantity of 5.5. That's consistent with my answer to (a).

(c) The industry appears to be competitive because marginal revenue is constant regardless of how much the firm produces. That indicates that the firm is likely a price taker.

If it is competitive, the industry isn't in long run equilibrium because the firm has non-zero profit.

4. (a)

| Q | FC | VC | AFC | AVC | ATC | MC |
|---|---|---|---|---|---|---|
| 0 | 100 | 0 | – | – | – | – |
| 1 | 100 | 50 | 100 | 50 | 150 | 50 |
| 2 | 100 | 70 | 50 | 35 | 85 | 20 |
| 3 | 100 | 90 | 33.33 | 30 | 63.33 | 20 |
| 4 | 100 | 140 | 25 | 35 | 60 | 50 |
| 5 | 100 | 200 | 20 | 40 | 60 | 60 |
| 6 | 100 | 360 | 16.67 | 60 | 76.67 | 160 |

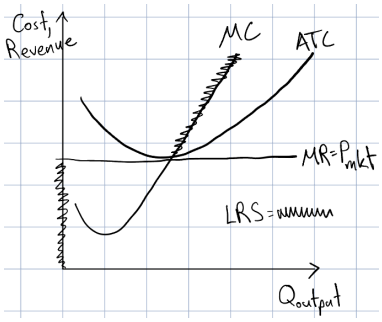
(b) If the firm shuts down, it makes no revenue, incurs no variable costs, and loses its fixed costs. In total, it loses \$100.

If the firm were to produce, it would produce the quantity at which $MC$ equals price. In this case, that's a quantity of one or four. If it produced one, it would make \$50, spend $\$100 + \$50 = \$150$,

and lose a total of \$100. If it produced four, it would make \$50·4 = \$200, spend \$100+\$140 = \$240, and lose a total of \$40. Therefore, producing four is better than producing one or shutting down. Shutting down is not a wise decision.

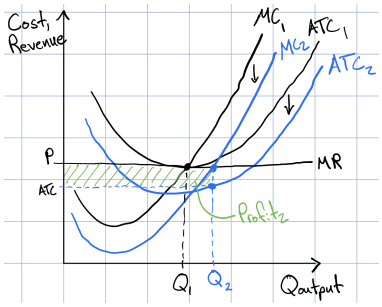
(c) The firm's losses are \$100 if it produces 1 case. It's not the best decision because if the firm produces 4 cases – the other quantity at which marginal cost equals marginal revenue – its losses are smaller, \$40. That's because the firm's marginal costs decline at first, so the firm's marginal costs are actually less than its marginal revenue for cases 2 & 3. As a result, producing cases 2 & 3 increases the company's profit.

5. (a) I show the long run supply curve. In order to show the short run supply curve, you would have to add in the AVC curve.



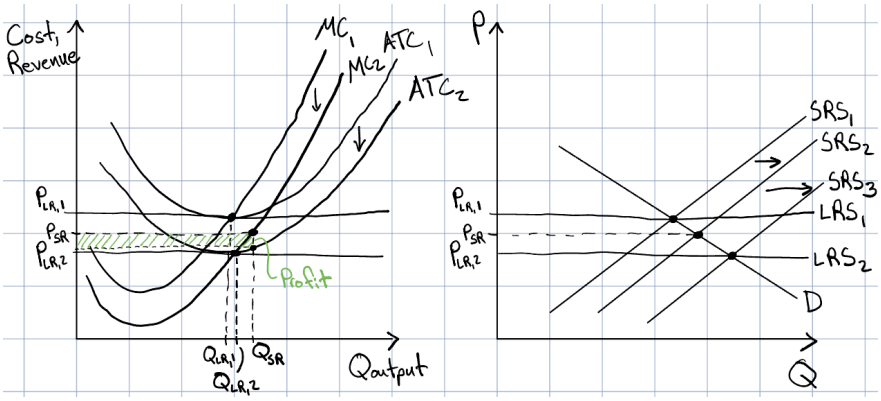

(b) Originally, Hi-Tech starts out in long-run equilibrium, so price equals marginal cost equals average total cost, and Hi-Tech's profits are zero.

After the invention, Hi-Tech's marginal cost and average total cost curves shift down. Because the market is competitive, Hi-Tech is a price taker. That means the change at Hi-Tech has no impact on the market price of a book. As a result, the intersection of price and Hi-Tech's new marginal cost curve is above Hi-Tech's new average total cost curve. That means Hi-Tech's profits become positive.

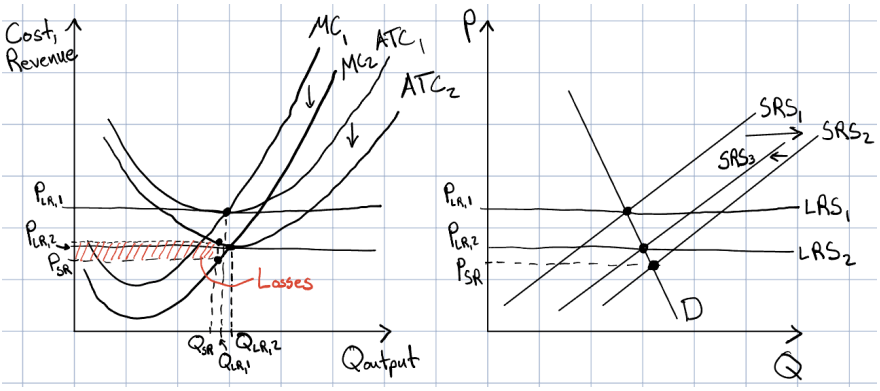



(c) In the long run, all firms will adopt the lower cost technology, so all of their marginal cost and average total cost curves will shift down and overlap with Hi-Tech's. Their marginal cost curves are their supply curves, so all of their supply curves are shifting to the right which means the short run market supply curve shifts to the right as well.

When short run market supply shifts right, it lowers the short run market price. Depending on the elasticity of demand, the new short run market price will be above or below the new long run market price. If it's above the new long run market price, firms will have positive profits, and new firms will enter the market. When new firms enter the market, it shifts the short run market supply curve further right until price hits its new long run equilibrium level. At that point, all firms' profits return to zero. That's the scenario I've drawn here.




If demand were less elastic, the new short run market price could be below the new long run market price. In that scenario, firms would have negative profits, and some would exit the market. When they exited the market, the short run market supply curve would shift back left until price rose to hit its new long run level. That's the scenario I've drawn here.




6. In a competitive market, all of a firm's output gets sold at the market price. That means marginal revenue equals average revenue equals the market price. Therefore, the average revenue and the price both equal \$10. Since the price is \$10, that means the firm sold \$500/\$10 = 50 units.

7. (a) Price equals average revenue which equals \$10. Therefore, profit is

$$\pi = (P - ATC) \cdot Q = (10 - 8) \cdot 100 = 200.$$

   (b) Since the firm is profit maximizing, marginal cost equals marginal revenue which equals the price of the good, \$10.

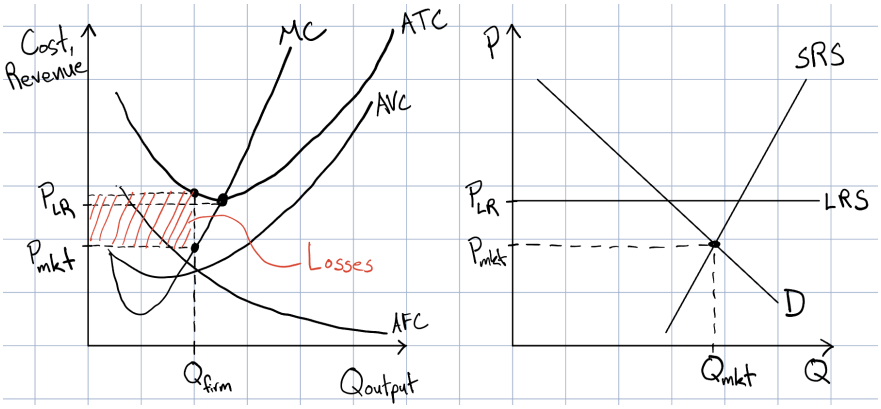
   (c) Average variable cost is

$$AVC = ATC - AFC = ATC - FC/Q = 8 - 200/100 = \$6.$$

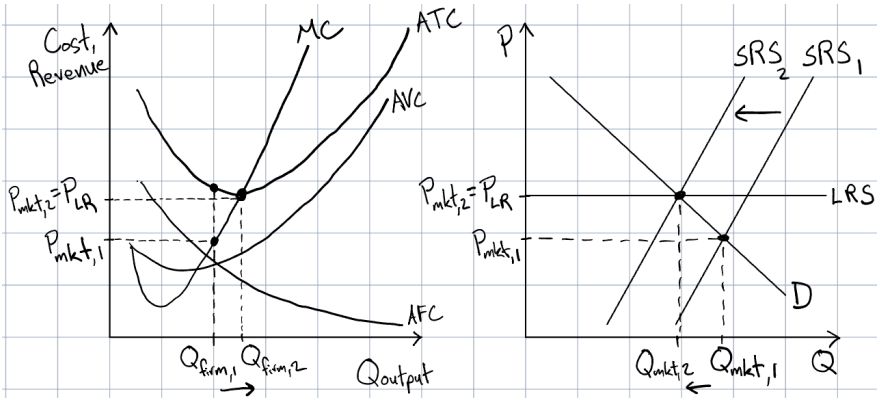
   (d) Since marginal cost is greater than average total cost, average total cost is greater than its minimum. That means the firm's efficient scale is less than 100 units.

8. (a) The firms are incurring losses, so price must be less than average total cost.
   The firms are still producing, so price must be greater than average variable cost.
   The firms are profit maximizers, so price must be equal to marginal cost.

   (b)




(c) Some firms exit the market because they are making losses. That decreases the number of suppliers which shifts short-run market supply left. When short-run market supply shifts left, price increases until it equals the minimum of average total cost and firm profits are zero.




The marginal cost and average total cost curves remain the same, so when price increases, firms start producing a quantity further right on the two curves. As a result, marginal cost increases, average total cost decreases, and the quantity supplied by each firm increases.

Total quantity supplied to the market decreases because of the leftward shift in short-run market supply.

9. (a)

| Q | MC | VC | TC | ATC |
|---|---|---|---|---|
| 1 pie | \$2 | \$2 | \$2 + \$9 = \$11 | \$11 1 = \$11 |
| 2 | 4 | 2 + 4 = 6 | 6 + 9 = 15 | 15/2 = 7.5 |
| 3 | 6 | 6 + 6 = 12 | 12 + 9 = 21 | 21/3 = 7 |
| 4 | 8 | 12 + 8 = 20 | 20 + 9 = 29 | 29/4 = 7.25 |
| 5 | 10 | 20 + 10 = 30 | 30 + 9 = 39 | 39/5 = 7.8 |
| 6 | 12 | 30 + 12 = 42 | 42 + 9 = 51 | 51/6 = 8.5 |

(b) If the price is \$11, that means the market is in equilibrium at $P = \$11$. At equilibrium, quantity demand equals quantity supplied, so we can read the quantity sold off of the demand schedule. 200 pies are sold.

Each producer will make 5 pies because 5 is the greatest quantity for which marginal cost is less than marginal revenue. Producing a sixth pie would cost them more than it would earn them.

There are $200/5 = 40$ producers.

Each producer has a profit of

$$\pi = (P - ATC) \cdot Q = (11 - 7.8) \cdot 5 = 16$$

(c) No, it's not because there are positive profits.

(d) In the long run, firms enter the market and drive down the price until profits are zero.
Price equals the minimum of average total cost which is \$7 (see part a.).
Each producer makes 3 pies because that's the quantity at which price equals the minimum of average total cost.
Based on the demand schedule, 600 pies are sold in the market.
There are $600/3 = 200$ producers.

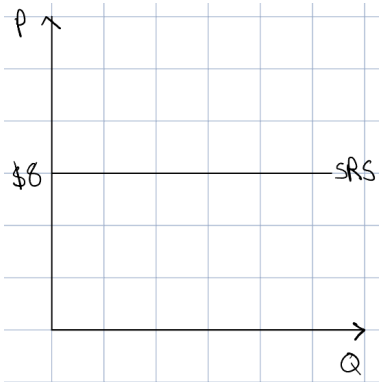
10. (a)

| Q | AVC | VC | MC | AFC | ATC |
|---|---|---|---|---|---|
| 1 | \$1 | $1 \cdot \$1 = \$1$ | $\$1 - \$0 = \$1$ | $\$16/1 = \$16$ | $\$1 + \$16 = \$17$ |
| 2 | 2 | $2 \cdot 2 = 4$ | $4 - 1 = 3$ | $16/2 = 8$ | $2 + 8 = 10$ |
| 3 | 3 | $3 \cdot 3 = 9$ | $9 - 4 = 5$ | $16/3 = 5.33$ | $3 + 5.33 = 8.33$ |
| 4 | 4 | $4 \cdot 4 = 16$ | $16 - 9 = 7$ | $16/4 = 4$ | $4 + 4 = 8$ |
| 5 | 5 | $5 \cdot 5 = 25$ | $25 - 16 = 9$ | $16/5 = 3.2$ | $5 + 3.2 = 8.2$ |
| 6 | 6 | $6 \cdot 6 = 36$ | $36 - 25 = 11$ | $16/6 = 2.67$ | $6 + 2.67 = 8.67$ |

(b) Each firm produces 5 units because 5 units is the greatest quantity for which marginal cost is less than marginal revenue. Producing a fifth unit increases the firms' profit because it brings in more money than it costs to produce. Producing any more than 5 units decreases the firms' profit because any additional unit brings in less money than it costs to produce.
Since there are 100 firms, the total quantity supplied in the market is $100 \cdot 5 = 500$ units.

(c) At an equilibrium price of \$10, firms produce 5 units at an average total cost of \$8.2. Since price is higher than average total cost, profits are positive. In the long run, that means firms will enter the market.
When firms enter the market, they will push the price down until profits are zero. That occurs when price equals the minimum of average total cost, \$8.
When the price falls, the quantity demanded will rise. That's just the law of demand.
When the price falls to \$8, each firm will produce 4 units – the highest quantity for which marginal cost is less than marginal revenue. Therefore, the quantity supplied by each firm will fall.

(d) Based on part (a), the minimum average total cost is approximately \$8.




11. (a) Fixed cost is the portion of total cost that doesn't depend on quantity, $FC = \$50$. Variable cost is the part of total cost that does depend on quantity, $VC = \frac{1}{2}q^2$. Average total cost is

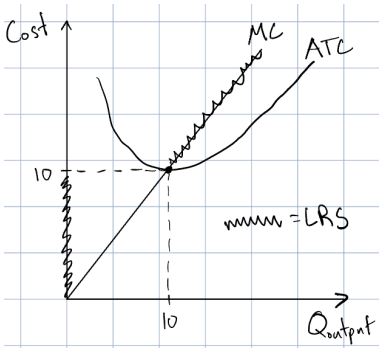
$$\begin{aligned} ATC &= \frac{TC}{q} \\ &= \frac{50 + \frac{1}{2}q^2}{q} \\ &= \frac{50}{q} + \frac{1}{2}q \end{aligned}$$

(b) You can solve this one graphically by finding the intersection of ATC and MC. You can solve it algebraically by setting ATC equal to MC. Or you can solve it with calculus by taking the derivative of ATC and setting it equal to zero. Solving graphically seems like the only not-so-great option to me. Here I solve it algebraically.

$$\begin{aligned} MC &= ATC \\ q &= \frac{50}{q} + \frac{1}{2}q \\ \frac{1}{2}q &= \frac{50}{q} \\ \frac{1}{2}q^2 &= 50 \\ q^2 &= 100 \\ q &= 10 \\ MC &= 10 \\ ATC &= 10 \end{aligned}$$

(c) The firm's long run supply curve is the marginal cost curve everywhere that $MC \geq ATC$. That means the firm's long run supply curve is

$$q = \begin{cases} P, & P \geq 10 \\ 0, & P < 10 \end{cases}$$



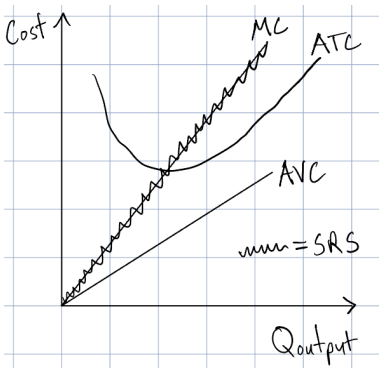

The firm's short run supply curve is the marginal cost curve everywhere that $MC \geq AVC$. Average variable cost is

$$\begin{aligned} AVC &= \frac{\frac{1}{2}q^2}{q} \\ &= \frac{1}{2}q. \end{aligned}$$

Therefore, $MC \geq AVC$ when

$$\begin{aligned} q &\geq \frac{1}{2}q \\ \frac{1}{2}q &\geq 0 \\ q &\geq 0. \end{aligned}$$

That means the firm's short run supply curve is $q = P$.




(d) There are nine firms in the market, so we just sum their individual short run supply curves to get

$$Q_S = 9q = 9P.$$

(e)

$$\begin{aligned} Q_S &= Q_D \\ 9P &= 120 - P \\ 10P &= 120 \\ P_{SR} &= 12 \\ Q_{SR} &= 9(12) \\ &= 108 \end{aligned}$$

(f) Each firm produces $q_{SR} = P_{SR} = 12$. Each firm's profit is

$$\begin{aligned} \text{Profit} &= TR - TC \\ &= Pq - (50 + \frac{1}{2}q^2) \\ &= 12(12) - (50 + \frac{1}{2}(12^2) \\ &= 144 - 122 \\ &= 22. \end{aligned}$$

Since profit is positive, firms have an incentive to enter the market.

(g) In the long run, the equilibrium price will be the minimum of ATC, so $P_{LR} = 10$. Market supply will be a horizontal line at $P_{LR} = 10$, so equilibrium quantity will be $Q_D$ when $P = 10$. That's $Q_{LR} = 120 - 10 = 110$.

(h) In the long run, firms still produce the quantity at which $P = MC$, so $q_{LR} = 10$. That means there are $110/10 = 11$ firms in the market.

## Non-Textbook Questions

1. (a) Market demand:

$$\begin{aligned} Q_D &= 10q_D \\ &= 10(40 - 2P) \\ &= 400 - 20P \end{aligned}$$

Individual firm supply:

$$\begin{aligned} P &= \frac{q_S}{4} \\ 4P &= q_S \end{aligned}$$

Market supply:

$$\begin{aligned} Q_S &= 5q_S \\ &= 5(4P) \\ &= 20P \end{aligned}$$

Short-run equilibrium:

$$\begin{aligned} Q_D &= Q_S \\ 400 - 20P &= 20P \\ 400 &= 40P \\ 10 &= P_{SR} \\ Q_{SR} &= 20(10) \\ &= 200 \end{aligned}$$

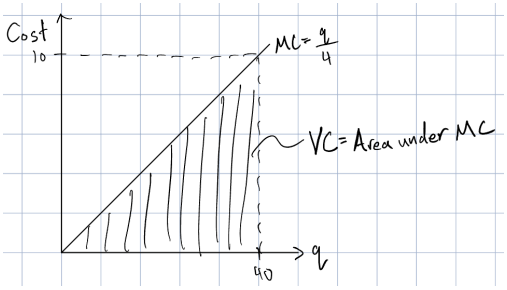
Firm-level production:

$$\begin{aligned} q_S &= 4(10) \\ &= 40 \end{aligned}$$

(b)

$$\begin{aligned} \pi &= TR - TC \\ &= P \cdot q - (FC + VC) \\ &= 10 \cdot 40 - (50 + VC) \\ &= 350 - VC \end{aligned}$$

We aren't given variable cost, but we know that variable cost is equal to the sum of marginal cost for every unit produced. Marginal cost is given by the height of the marginal cost curve. To sum marginal cost from zero to forty, we can take the area under the curve. (In calculus terms, $VC = \int MC\, dq$.)




$$\begin{aligned} VC &= \frac{1}{2}(10)(40) \\ &= 200 \end{aligned}$$

Therefore,

$$\begin{aligned} \pi &= 350 - 200 \\ &= 150. \end{aligned}$$

Since profits are positive, firms will enter the market in the long run.

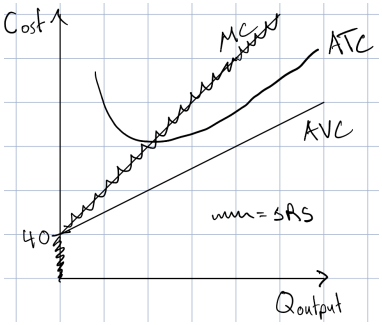
2. (a) A firm will shut down at any price where its marginal cost curve is below its average variable cost curve. The quantity at which the marginal cost curve goes below the average variable cost curve is:

$$\begin{aligned} MC &\leq AVC \\ MC &\leq \frac{VC}{q} \\ 40 + q &\leq \frac{40q + \frac{q^2}{2}}{q} \\ 40 + q &\leq 40 + \frac{q}{2} \\ \frac{q}{2} &\leq 0 \\ q &\leq 0 \end{aligned}$$

The corresponding price is:

$$\begin{aligned} MC &= 40 + 0 \\ MC &= 40 \\ P &= 40 \end{aligned}$$

Therefore, firms will shut down in the short run when the price falls below \$40.




(b) Individual firm supply for all prices greater than 40:

$$\begin{aligned} P &= 40 + q \\ P - 40 &= q \end{aligned}$$

Market supply:

$$\begin{aligned} Q_S &= 12q \\ &= 12(P - 40) \\ &= 12P - 480 \end{aligned}$$

Short-run equilibrium price:

$$\begin{aligned} 12P - 480 &= 360 - 2P \\ 14P &= 840 \\ P &= 60 \end{aligned}$$

3. (a) The long-run equilibrium price will be the minimum of the firms' average total cost curves. The

minimum occurs at the quantity where marginal cost equals average total cost:

$$\begin{aligned} MC &= ATC \\ 20 + 2q &= 20 + q + \frac{144}{q} \\ q &= \frac{144}{q} \\ q^2 &= 144 \\ q &= 12 \end{aligned}$$

Thus, the long-run equilibrium price is

$$\begin{aligned} P &= 20 + 2(12) \\ &= 44 \end{aligned}$$

(b) Individual firms will produce the quantity at which price equals marginal cost. That's

$$\begin{aligned} 44 &= 20 + 2q \\ 12 &= q. \end{aligned}$$

(c) If the equilibrium price is \$44, then the equilibrium quantity is

$$\begin{aligned} Q &= 2,488 - 2(44) \\ &= 2,400. \end{aligned}$$

If each firm produces 12 units, there must be $2,400\ 12 = 200$ producers.